basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

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ANNUAL NATIONAL ASSESSMENT 2015
GRADE 7 MATHEMATICS TEST

MARKS: 100

MARKS	

TIME: 2 hours

PROVINCE _____

DISTRICT _____

CIRCUIT _____

SCHOOL _____

EMIS NUMBER (9 digits)

CLASS (e.g. 7A) _____

SURNAME _____

NAME _____

GENDER (✓)

BOY		GIRL	

DATE OF BIRTH

C	C	Y	Y	M	M	D	D

This test consists of 14 pages, excluding the cover page.

Instructions to the learner

1. Read all the instructions carefully.
2. Question 1 consists of 10 multiple-choice questions. You must circle the letter of the correct answer.
3. Answer questions 2 to 8 in the spaces provided.
4. All working must be shown.
5. The diagrams are not drawn to scale.
6. The test is out of 100 marks.
7. The test duration is 2 hours.
8. The teacher will lead you through the practice question before you start the test.
9. You may use a calculator unless otherwise stated.

Practice question.

Circle the letter of the correct answer.

1. The next number in the number sequence $1\ ;\ 3\ ;\ 5\ ;\ 7\ ;$ ... is …

   A 8

   B 12

   C 16

   D 9

Your answer is correct if you circled D.

The test starts on the next page.

QUESTION 1

1.1 The lowest common multiple of 5 and 7, is …

A 5

B 35

C 12

D 7 (1)

1.2 In $6x + 2$, the variable is …

A $6x$

B $6x + 2$

C x

D 6 (1)

1.3 32 written as a product of its prime factors is …

A 1×32

B 2×16

C $2 \times 2 \times 2 \times 2 \times 2$

D $2 \times 4 \times 4$ (1)

1.4 What percentage is $1\,200$ of $5\,000$?

A 24%

B 50%

C 38%

D 12% (1)

1.5 What is the value of $3 - \frac{k}{2}$ if $k = 4$?

A 4

B 1

C 2

D -2 (1)

1.6 A quadrilateral in which all 4 sides are equal, is called a …

A rectangle.

B parallelogram.

C rhombus.

D kite. (1)

1.7 A set of points with a definite starting-point and no definite end-point is called a …

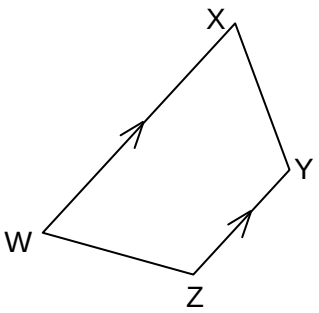
A line segment.

B ray.

C straight line.

D perpendicular line. (1)

1.8



In the above figure, WX and ZY are…

A perpendicular line segments.

B parallel line segments.

C intersecting line segments.

D vertical line segments. (1)

1.9 The perimeter of a regular pentagon with one side equal to 5 cm is …

A 25 cm

B 5 cm

C 35 cm

D 30 cm (1)

1.10

x	1	2	3	4
y	4	5	6	7

The relationship between x and y is …

A $y = 5 \times x$

B $y = 3 \times x$

C $y = x + 4$

D $y = x + 3$ (1)

[10]

QUESTION 2

2.1 Calculate without using a calculator. Show the calculation steps where needed.

2.1.1 $1\,643\,884 + 262\,206$

_____

_____

_____

_____ (2)

2.1.2 $6\,517 \div 31$

(4)

2.1.3 315×236

(4)

2.1.4 $2^3 \times 1^4$

(3)

2.1.5 $\sqrt{144} + 6^2$

(3)

2.1.6 $\frac{3}{5} \times \frac{7}{6}$

_____

_____

_____

_____ (3)

2.1.7 $0{,}012 \div 4$

_____

_____ (1)

2.1.8 $100 - 12 \div (8 + 4)$

_____

_____

_____ (3)

2.2 Write $1{,}6$ as an improper fraction in the simplest form.

_____

_____ (2)

2.3 A group of 25 Grade 7 learners watched a soccer match. The ratio of the number of boys to the number of girls was $2 : 3$. How many boys were there?

_____

_____

_____

_____

_____ (3)

2.4 Calculate 10% of $R15{,}00$.

_____

_____

_____ (3)

2.5 John travelled by car at an average speed of 80 kilometres per hour for $4\frac{1}{2}$ hours. Calculate how far he travelled.

_____

_____

_____

_____ (3)

[34]

QUESTION 3

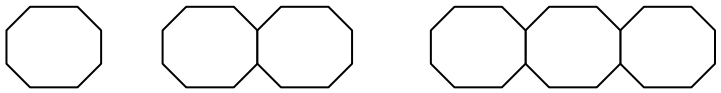
3.1 3.1.1 Fill in the missing numbers in the following number sequence:

1 ; 4 ; 9 ; _____ ; 25 ; _____. (2)

3.1.2 Are the numbers in the above number sequence prime numbers, square numbers or cubic numbers?

_____ (1)

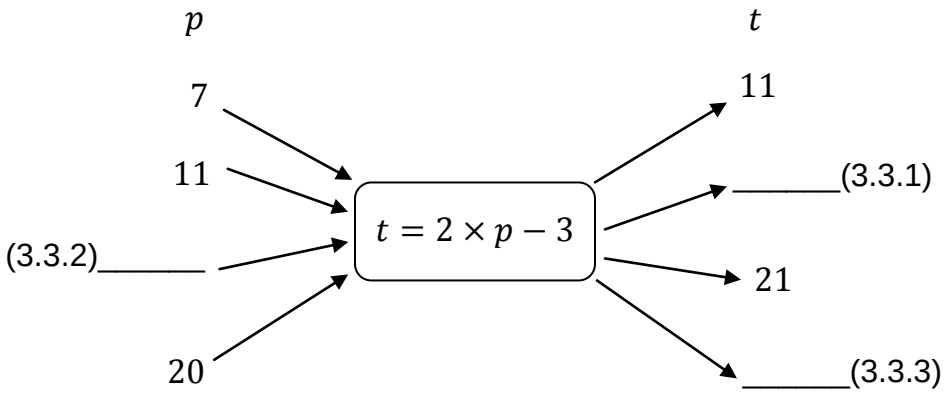
3.2 Study the following diagram pattern of regular octagons and then complete the table.



Number of polygons	1	2	3	4	n
Number of sides	8	15	22		

(3)

3.3 Use the given rule to fill in the missing values in the flow diagram below.



(3)

3.4 Calculate the value of $c^3 + 12$ if $c = 3$.

_____

_____

_____

_____ (3)

3.5 Solve the following equations by inspection:

3.5.1 $\frac{k}{3} = 9$

_____ (1)

3.5.2 $y - 17 = 13$

_____ (1)

3.6 3.6.1 Write an algebraic expression for the following:

x increased by five

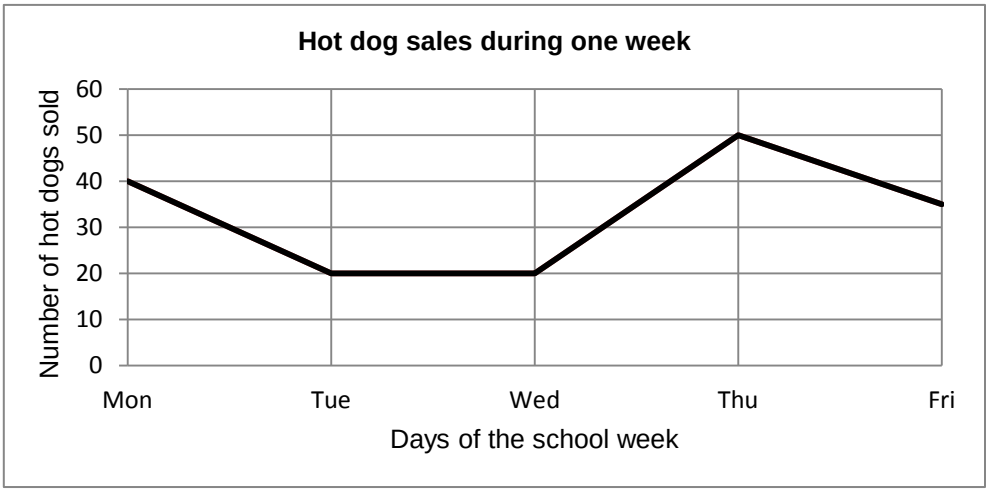
_____ (1)

3.6.2 Write a number sentence for the following:

Two times a certain number p decreased by eight is equal to twelve.

_____ (3)

3.7 The given graph illustrates the number of hot dogs sold on different school days during one week.



3.7.1 How many hot dogs were sold on Thursday?

_____ (1)

3.7.2 On which days did the sales remain constant?

_____ (1)

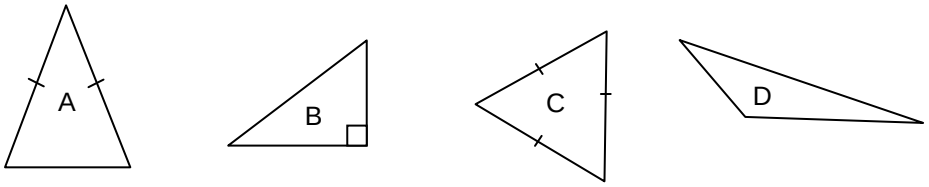
3.7.3 Calculate the difference between the number of hot dogs sold on Monday and on Thursday.

_____ (2)

[22]

QUESTION 4

4.1 Study the triangles marked A, B, C and D.

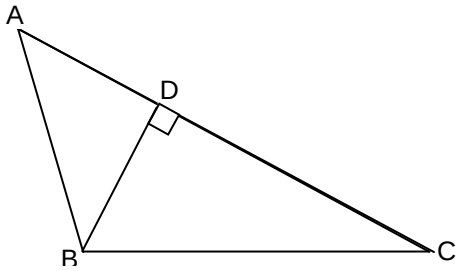


Complete the table below by filling in A, B, C or D.

Equilateral triangle	Right-angled triangle	Obtuse-angled triangle

(3)

4.2 Study the diagram below and answer the questions that follow.



4.2.1 What kind of angle is $\hat{C}$?

_____ (1)

4.2.2 What kind of angle is $A\hat{B}C$?

_____ (1)

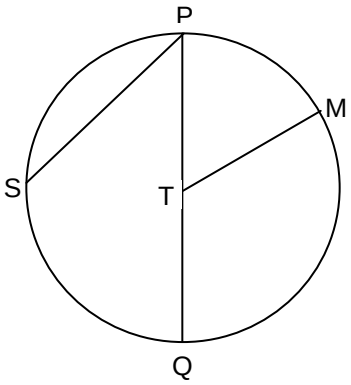
4.2.3 What kind of angle is $B\hat{D}C$?

_____ (1)

4.2.4 Name a pair of perpendicular line segments.

_____ (1)

4.3 T is the centre of the circle. Name the parts of the circle that are listed.



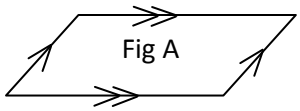
4.3.1 PQ is called _____.

4.3.2 TM is called _____.

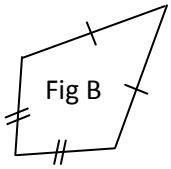
4.3.3 PS is called _____.

(3)

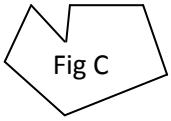
4.4 . Identify each of the figures below:



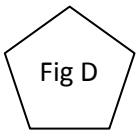
4.4.1 Figure A is a _____



4.4.2 Figure B is a _____



4.4.3 Figure C is a _____



4.4.4 Figure D is a _____

(4)

[14]

QUESTION 5

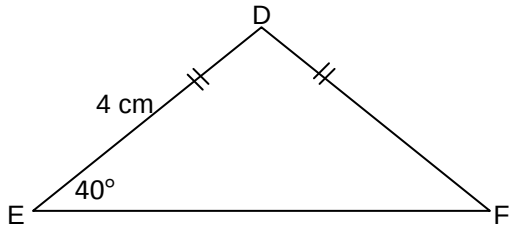
5.1 In an equilateral triangle PQR, $\mathrm{PQ} = 23$ mm

Complete:

5.1.1 $\mathrm{QR} = \mathrm{PR} =$ _____ mm (1)

5.1.2 $\widehat{\mathrm{P}} = \widehat{\mathrm{Q}} = \widehat{\mathrm{R}} =$ _____° (1)

5.2 Study the diagram below and then, answer the questions that follow.



5.2.1 What is the length of DF?

_____ (1)

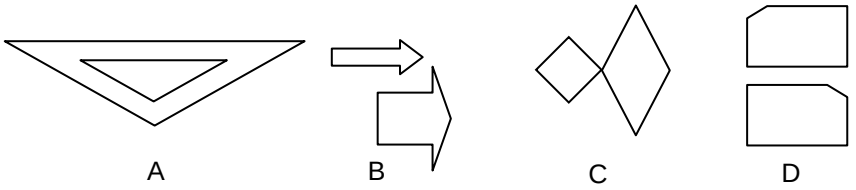
5.2.2 If $\hat{\mathrm{E}} = 40°$, what is the size of $\hat{\mathrm{F}}$?

_____ (1)

[4]

QUESTION 6

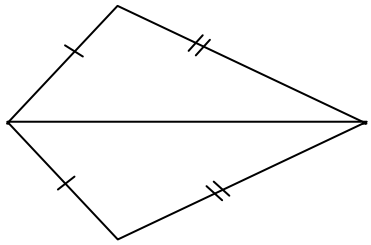
6.1 Study the pairs of diagrams below and complete the statements by filling in A or B or C or D in the spaces provided.



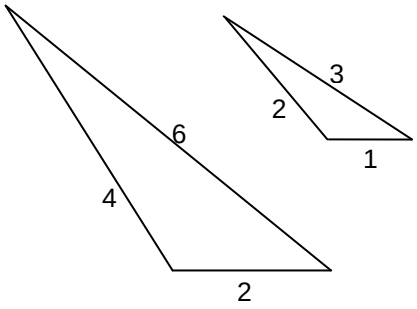
6.1.1 The 2 shapes in _____ are congruent. (1)

6.1.2 The 2 shapes in _____ are similar. (1)

6.2 Write down whether each pair of triangles given below is congruent, similar or neither.



Pair 1



Pair 2

6.2.1 Triangles in pair 1 are _____. (1)

6.2.2 Triangles in pair 2 are _____. (1)

[4]

QUESTION 7

7.1 Calculate the perimeter of the rectangle if the length is 11 cm and the breadth is 5 cm.

_____

_____

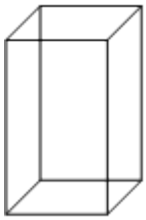
_____

_____ (3)

7.2 Write down the formula for the surface area of a cube.

_____ (1)

7.3 In the rectangular prism below, the length $= 5$ cm, the breadth $= 3$ cm and the height $= 10$cm.



Calculate the volume of the rectangular prism.

_____

_____

_____

_____ (3)

[7]

QUESTION 8

8.1 A mother is twice as old as her daughter. How old is the daughter if the mother is 64 years old?

_____

_____ (2)

8.2 The product of two prime numbers is 10. What is the difference between the two prime numbers?

_____

_____ (3)

[5]

TOTAL: 100